

REQUEST FOR INFORMATION **W911SR-26-X-0002**

This Request for Information (RFI) is issued in accordance with Federal Acquisition Regulation (FAR) 15.201(c)(7) for the purpose of conducting market research. This RFI is not a solicitation, request for proposal, or request for prototype proposals. No contract will be awarded from this announcement. The Government will not reimburse respondents for any costs incurred in preparing or submitting a response to this RFI. No telephone calls, office calls, or one-on-one meetings will be accepted or conducted as a result of this notice.

All responses, questions, and comments shall be submitted via email to:

- **Elizabeth Nordell** – elizabeth.m.nordell.civ@army.mil
- **Sarah Richardson** – sarah.h.richardson3.civ@army.mil

Responses are due **no later than May 15, 2026 at 10:00 AM Eastern**. All submissions shall be provided in Microsoft Word, Microsoft PowerPoint, or PDF compatible format. Proprietary information must be clearly marked and separated.

1.0 PURPOSE

The U.S. Army Combat Capabilities Development Command - Chemical Biological Center (DEVCOM CBC) Chemical Biological Applications and Risk Reductions (CBARR) business unit is searching for potential sources of equipment or solutions to rapidly render bulk volumes of toxic industrial chemicals (TICs) safer by reducing their toxicity or disseminability. This RFI seeks to identify existing or adaptable candidate technologies and strategies to reduce the operational risk associated with selected TICs/Chemicals of Security Concern (COSC).

The Government intends to use the information received to refine requirements, assess technology readiness, and inform future acquisition planning. This RFI does not commit the Government to any future procurement.

2.0 TECHNICAL DESCRIPTION

DEVCOM CBC is seeking information concerning rendering portable, transportable, and/or bulk volumes of toxic industrial chemicals (TICs) safe(r) to reduce the operational risk associated with selected TICs/Chemicals of Security Concern (COSC) using any of the following methods:

- Neutralization
- Reduction in toxicity
- Reduction in disseminability
- Other reactive or passive technological applications, strategies, or approaches

Below is a list of the Chemicals of Security Concern:

GC IAG Chemical of Security Concern	CASRN
Acrolein	107-02-8
Ammonia (anhydrous)	7664-41-7
Arsenic Trichloride	7784-34-1
Arsine (SA)	7784-42-1
Boron Trifluoride	7637-07-2
Bromine	7726-95-6
Chlorine	7782-50-5
Cyanogen Chloride (CK)	506-77-4
Ethylene	74-85-1
Fluorine	7782-41-4
Hydrogen Chloride	7647-01-0
Hydrogen Fluoride/hydrofluoric acid (>50%)	7664-39-3
Hydrogen Sulfide	7783-06-4
Phosgene (CG)	75-44-5
Phosphorus Oxychloride	10025-87-3
Phosphorus Trichloride	7719-12-2

DEVCOM CBC is primarily seeking information from vendors with technically mature equipment or solutions that meet the technical performance goals of system size, reactant ratios and proof of scalability listed in this section of the RFI and the below/attached “Chemical of Security Concern (CoSC) Risk Reduction Goals”.

DEVCOM CBC requests vendors provide proof of a proposed system’s capability to result in a reduction in toxicity or a reduction in disseminability as defined in the below/attached “Chemical of Security Concern (CoSC) Risk Reduction Goals”.

THIS IS A REQUEST FOR INFORMATION ONLY

All submissions become Government property and will not be returned. Respondents shall provide a brief description of how they plan to develop or advance existing technologies/methodologies or materials for the identified applications as well as any anticipated technical parameters of their product (or conceptual product). Validation can include historical data, literature data, data from a relationship with current Department of War (DoW) researchers, or data from previous governmental interagency programs.

3.0 REQUESTED INFORMATION

Respondents shall address the following items in Arial 12 point font, in no more than 15 pages total. Responses shall be submitted in Microsoft Word-compatible or PDF format.

Please ensure your response addresses the prompts below in addition to the other items listed in this section

- Describe your company's existing or adaptable candidate technologies and strategies to render bulk volumes of toxic industrial chemicals (TICs) safer for transportability.
- As experts in your field, provide additional information you believe we should know in support of this RFI to assist the Government in its market research and acquisition planning.

3.1 Business Information

Provide the following business information:

- Company Name
- Address
- Point of Contact
 - POC Phone Number
 - POC Email Address
- Business Size (Large or Small), using the size standard under NAICS 541512 unless another NAICS is justified (see SBA Size Standards: <https://www.sba.gov/document/support-table-size-standards>)
- Small Business Representation (8(a), SDB, WOSB, HUBZone, VOSB, SDVOSB, etc.)

4.2 Technical Capability

Address your ability to meet the technical requirements in Section 2.0 and the attached "Chemicals of Security Concern (CoSC) Risk Reduction Goals/Technical Requirements".

- Identify any gaps your company has related to technical capability or the ability to manufacture a deployable product.
- Provide the Government insight into what would be required to bridge the gap between your current product and an end-user-ready system that meets the specifications in Section 2.0.
- Provide the Government with unit cost for higher TRL technologies or rough order of magnitude (ROM) cost for lower TRL technologies as outlined in Section 2.0.

Additionally, your response should include information regarding the following characteristics:

- Ease of operation
- Stability, Longevity and Availability
- Ease of Reversibility
- Availability of data and source information

Pricing will not be construed to be an offer and is simply to provide the Government insight to roughly gauge current and short term market realities for this technology. Responses shall not be above the level of Unclassified. Proprietary Information should be marked as such.

4.3 Relevant Past Performance

Provide up to three examples of similar work, including:

- Contract number
- Prime or subcontractor role
- Dollar value
- Period of performance
- Relevance to this requirement

4.4 Accounting System

State whether you possess—or can and are willing to obtain—a DCAA-approved accounting system. Address whether a cost-reimbursable contract type would be required or useful to mature your product into an end-user-ready capability.

4.5 Purchasing System

State whether you or a potential teaming partner possess a purchasing system compliant with DFARS 252.244-7001(c).

4.6 Facility Clearance

State what, if any, facility clearance level your company currently possesses (e.g., Confidential, Secret, Top Secret).

4.7 Additional Information

Provide any additional information relevant to DEVCOM CBC's market research.

5.0 INDUSTRY ENGAGEMENT

DEVCOM CBC will not conduct one-on-one meetings during the RFI period. Questions will be collected and addressed collectively during Industry Day. Participation in this RFI is not required for participation in any future solicitation.

Chemicals of Security Concern (CoSC) Risk Reduction Goals

Technical Solution Performance Goals: The performance criteria of system size, reactant ratio (reactive candidates), and scalability are applicable to every case; however, any combination of the other three performance criteria (reduction in toxicity, reduction of the ability to disseminate, reduction in volatility) that achieves a 50% reduction in overall hazard, whether through a single criterion or combination, is acceptable.

Performance Goals. The following are the criteria for technical solution performance:

1. System Size:

- **Definitions**
 - Large Bulk Liquid Containers: 1000L totes.
 - Large Bulk Gas Containers: 100lb compressed gas cylinders.
- **Restrictions**
 - If the proposed solution is readily available at a Chemical Production Facility, this criterion will not apply.
 - System footprint is not restricted
- **Goals**
 - The proposed technical solution's mass is:
 1. Threshold: Less than the mass of the target CoSC.
 1. Preferred: ≤ 500 kg.
 2. Ceiling Value: Large bulk container solutions have a maximum mass (ceiling weight of equipment) of 3,000kg.
 2. Objective: ≤ 36 kg.

2. Reactant Ratio:

- **Definitions**
 - Reactant: A substance added to the CoSC to alter its chemical properties.
- **Restrictions**
 - If the technical solution does not require the addition of a reactant, this criterion will not apply.
- **Goals**
 - If reactants are used, the reactant-to-CoSC ratio is:
 1. Threshold: $\leq 2:1$.
 2. Objective: $<1:1$.

3. Indication/Proof of Scalability:

- **Definitions**
 - Sub-scale: Less than 100mL (minimum required for analytical laboratory testing).
 - Laboratory Scale: 100-1000mL.
 - Field-Testing Scale: 1-10L.
 - Bulk Container Scale: 1000L totes (liquid) and 100lb compressed gas cylinders (gas).
- **Restrictions**
 - N/A
- **Goals**
 - The approach meets, or is expected to meet, the applicable performance goals at four scales: subscale, laboratory scale, field-testing scale, and bulk container scale.

Reducing Toxicity (Evaluated Only if the Solution Does Not Focus on Disseminability)

1. Reduction in Toxicity

- **Definitions**
 - Toxicity: Measured based on the concentration of the CoSC present and based on the following exposure limits.
 1. IDLH – immediately dangerous to life and health
 2. PEL -the OSHA Permissible Exposure Limit. If the PEL is not published, another federal exposure limits may be used.
- **Restrictions**
 - Set-up time is not included in the duration evaluation.
 - The toxicity of the resulting reactants will only be considered if their exposure limits are equal to or higher than the original CoSC.
- **Goals**
 - The technical solution will reduce the CoSC's toxicity:
 1. Threshold: By 50%.
 2. Objective: below the PEL.

AND

- The reduction in toxicity will occur within:
 1. Threshold: ≤ 7 days.
 2. Objective: ≤ 2 hours.

Performance Goals for Reducing Disseminability (Evaluated Only if the Solution Does Not Focus on Toxicity)

To pass the performance goals for reducing disseminability, only one of the following criteria must be met.

1. Reduction in Disseminability

- **Definitions**
 - Disseminability -the potential to be aerosolized and/or weaponized/ released for harmful purposes.
- **Restrictions**
 - Evaluation of a solution's disseminability will be measured qualitatively to assess its weaponization potential using CBARR's subject matter expertise.
- **Goals**
 - The technical solution will reduce the CoSC's disseminability by:
 1. Threshold: Transforming its final state into a non-aerosolizable liquid or gel.
 2. Objective: Transforming its final state into a solid.

2. Reduction in Vapor Pressure

- **Definitions**
 - Vapor pressure will be measured using a standard method or evaluated based on published values.
- **Restrictions**
 - N/A
- **Goals**

- The technical solution will reduce the CoSC's vapor pressure by:
 1. Threshold: 50% reduction from the initial value.
 - a. Alternative Approach: Sealing the surface of the CoSC to prevent vapor escape will meet the threshold.
 2. Objective: Achieving a vapor pressure of ≤ 0.0007 mmHg.

3. Reduction in Volatility

- **Definitions**
 - Volatility will be measured at 25°C.
- **Restrictions**
 - N/A
- **Goals**
 - The technical solution will reduce the CoSC's volatility by:
 1. Threshold: 50% reduction from the initial value.
 - a. Alternative Approach: Sealing the surface/encapsulation of the CoSC to prevent vapor escape will meet the threshold.
 2. Objective: Achieving a volatility of ≤ 10.5 mg/m³.